

1. What is a compass?

OV-SVD $\rightarrow$ rank-2 dial

- For head h at layer ℓ , factor $W_{OV} = W_V W_O = U \Sigma V^\top$.

- A **compass plane** is two right-singular vectors (u_i, u_j) . Inject along it:

$$v(\theta, \alpha) = \alpha \sigma_i \cos \theta u_i + \alpha \sigma_j \sin \theta u_j.$$

- If the plane is causal, the logit-difference swings **sinusoidally** in θ and **linearly** in α ; if it is just correlational, random planes and label permutations reproduce the same numbers.
- The rest of this deck shows that (i) the sinusoidal + linear response really holds, (ii) compasses exist in every model we looked at, and (iii) they can drive a debiasing stack.